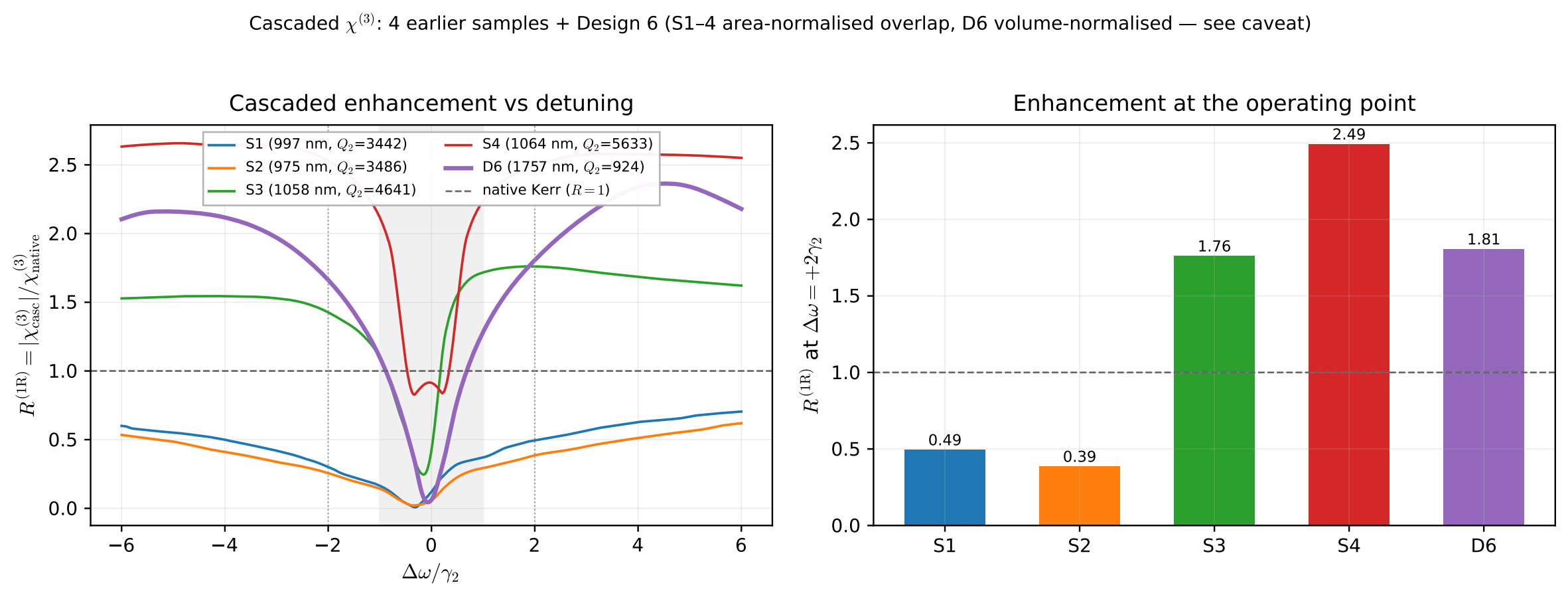
Cascaded $\chi^{(3)}$: 4 earlier samples + Design 6 (S1–4 area-normalised overlap, D6 volume-normalised — see caveat)
Cascaded enhancement vs detuning
S1 (997 nm, Q_2=3442)
S2 (975 nm, Q_2=3486)
S3 (1058 nm, Q_2=4641)
S4 (1064 nm, Q_2=5633)
D6 (1757 nm, Q_2=924)
native Kerr ($R=1$)
$R^{(1R)} = |\chi^{(3)}_{casc}| / \chi^{(3)}_{native}$
$\Delta\omega/\gamma_2$
Enhancement at the operating point
$R^{(1R)}$ at $\Delta\omega = +2\gamma_2$
0.49
0.39
1.76
2.49
1.81
S1
S2
S3
S4
D6